CHAPTER 2 DETAILED SUMMARY

SECTION 2.1 The Rectangular Coordinate System and Graphing Utilities

KEY CONCEPTS

The **distance** between two points (x_1, y_1) and (x_2, y_2) in a rectangular coordinate system is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The **midpoint** between the points is given by

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Graphing an equation:
To graph an equation in a rectangular coordinate system, plot several solutions to the equation to form a general outline of the curve. Then connect the points to form a smooth line or curve.

Determining *x*- and *y*-intercepts:
To find an x-intercept $(a, 0)$ of the graph of an equation, substitute 0 for y and solve for x.

To find a y-intercept $(0, b)$ of the graph of an equation, substitute 0 for x and solve for y.

EXAMPLES

Example 1:
Given $(-4, -5)$ and $(6, -1)$, find the distance between the points and the midpoint of the line segment between the points.

$$d = \sqrt{[6 - (-4)]^2 + [-1 - (-5)]^2}$$

$$d = \sqrt{(10)^2 + (4)^2} = \sqrt{116} = 2\sqrt{29}$$

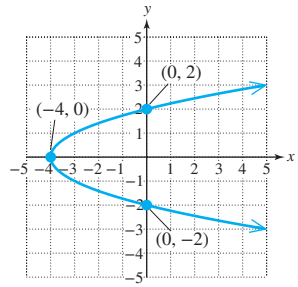
$$M = \left(\frac{-4 + 6}{2}, \frac{-5 + (-1)}{2}\right) = (1, -3)$$

Example 2:
Graph $x = y^2 - 4$.

x	5	0	−3	−4	−3	0	5
y	−3	−2	−1	0	1	2	3

***x*-intercept:**
$x = y^2 - 4$
$x = (0)^2 - 4$
$x = -4$

***y*-intercepts:**
$x = y^2 - 4$
$0 = y^2 - 4$
$y = 2$ or $y = -2$



SECTION 2.2 Circles

KEY CONCEPTS

A **circle** is the set of all points in a plane that are equidistant from a fixed point called the **center.** The fixed distance between the center and any point on the circle is called the **radius** of the circle.

The **standard form** of an equation of a circle with radius r and center (h, k) is

$$(x - h)^2 + (y - k)^2 = r^2$$

An equation of a circle written in the form $x^2 + y^2 + Ax + By + C = 0$ is called the **general form** of an equation of a circle.

Writing an equation in standard form:
By completing the square, we can write an equation of a circle given in general form as an equation in standard form. The purpose is to identify the center and radius from the standard form.

EXAMPLES

Example 1:
Write the standard form of an equation of the circle whose center is $\left(3, -\frac{1}{2}\right)$ and whose radius is 5.

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - 3)^2 + \left[y - \left(-\frac{1}{2}\right)\right]^2 = (5)^2$$

$$(x - 3)^2 + \left(y + \frac{1}{2}\right)^2 = 25$$

Example 2:
Write the equation of the circle in standard form. Then identify the center and radius.

$$x^2 + y^2 - 12x + 8y + 5 = 0$$

$$x^2 - 12x + 36 + y^2 + 8y + 16 = -5 + 36 + 16$$

$$(x - 6)^2 + (y + 4)^2 = 47$$

The center is $(6, -4)$ and the radius is $\sqrt{47}$.

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Not all equations of the form $x^2 + y^2 + Ax + By + C = 0$ represent the graph of a circle. Completing the square results in an equation of the form $(x - h)^2 + (y - k)^2 = c$ where c is a constant.

- If $c > 0$, then the graph will be a circle with radius $r = \sqrt{c}$.
- If $c = 0$, then the graph will be a single point (h, k). The solution set is $\{(h, k)\}$. (Degenerate case)
- If $c < 0$, then the solution set is the empty set $\{\ \}$. (Degenerate case)

Example 3:

$$x^2 + y^2 - 8x + 2y + 17 = 0$$
$$x^2 - 8x + 16 + y^2 + 2y + 1 = -17 + 16 + 1$$
$$(x - 4)^2 + (y + 1)^2 = 0$$

The sum of two squares will equal zero only if each individual term is zero. Therefore, $x = 4$ and $y = -1$.

The solution set is $\{(4, -1)\}$.

SECTION 2.3 Functions and Relations

KEY CONCEPTS

A set of ordered pairs (x, y) is called a **relation** in x and y. The set of x values is the **domain** of the relation, and the set of y values is the **range** of the relation.

Given a relation in x and y, we say that ***y* is a function of *x*** if for each value of x in the domain, there is exactly one value of y in the range.

Vertical line test:
The graph of a relation defines y as a function of x if no vertical line intersects the graph in more than one point.

Evaluating a function for different values of *x*:
A function may be evaluated at different values of x by using substitution.

Determining *x*- and *y*-intercepts:
Given a function defined by $y = f(x)$,

- The x-intercept(s) are the real solutions to $f(x) = 0$.
- The y-intercept is given by $f(0)$.

EXAMPLES

Example 1:
Given the relation $\{(8, 2), (3, 2), (9, 5)\}$,

The domain is $\{8, 3, 9\}$.
The range is $\{2, 5\}$.

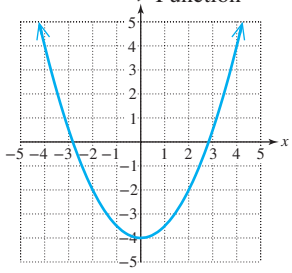
Example 2:

The relation $\{(4, 6), (3, 5), (4, 1)$ is *not* a function.
(same x over the first two 4's; different y over 6 and 1)

The relation $\{(3, 5), (2, 1), (9, 0)\}$ is a function because no two ordered pairs have the same x value but different y values.

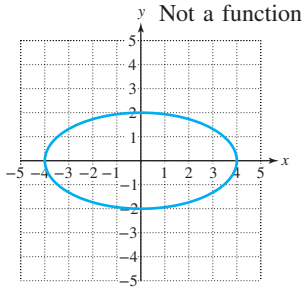
Example 3:

y Function



Domain: $(-\infty, \infty)$
Range: $[-4, \infty)$

y Not a function



Domain: $[-4, 4]$
Range: $[-2, 2]$

Example 4:
Given $f(x) = 2x^2 + 3x$,

$$f(2) = 2(2)^2 + 3(2) = 14$$
$$f(x + 4) = 2(x + 4)^2 + 3(x + 4)$$
$$= 2(x^2 + 8x + 16) + 3x + 12$$
$$= 2x^2 + 19x + 44$$

Example 5:
Given $f(x) = |x| - 2$,

- To find the x-intercept(s), substitute 0 for $f(x)$:
  $0 = |x| - 2$
  $x = 2$ or $x = -2$
  The x-intercepts are $(2, 0), (-2, 0)$.
- To find the y-intercept, evaluate $f(0)$.
  $f(0) = |0| - 2 = -2$ The y-intercept is $(0, -2)$.

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Determining domain from $y = f(x)$:
Given $y = f(x)$, the domain of f is the set of real numbers x that when substituted into the function produce a real number. This excludes

- Values of x that make the denominator zero.
- Values of x that make a radicand negative within an even-indexed root.

Function values and the domain and range of a function can be estimated from the graph of the function.

Example 6:

- Given $f(x) = \dfrac{x+5}{x-3}$, the domain is $(-\infty, 3) \cup (3, \infty)$.
- Given $g(x) = \sqrt{x-3}$, the domain is $[3, \infty)$.

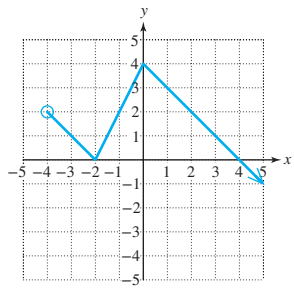
Example 7:
From the graph, find

a. the domain

b. the range

c. $f(3)$

d. the values of x for which $f(x) = 2$



a. $(-4, \infty)$

b. $(-\infty, 4]$

c. $f(3) = 1$

d. $x = -1, x = 2$

SECTION 2.4 Linear Equations in Two Variables and Linear Functions

KEY CONCEPTS

Let A, B, and C represent real numbers, where A and B are not both zero. A **linear equation** in the variables x and y is an equation that can be written as $Ax + By = C$.

The graph of a linear equation is a line.

- A horizontal line has an equation of the form $y = k$, where k is a constant real number.
- A vertical line has an equation of the form $x = k$, where k is a constant real number.

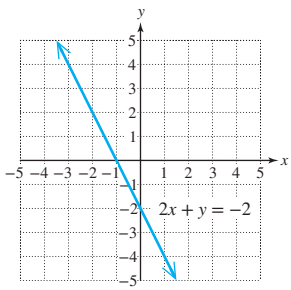
The slope of a line passing through the distinct points (x_1, y_1) and (x_2, y_2) is given by

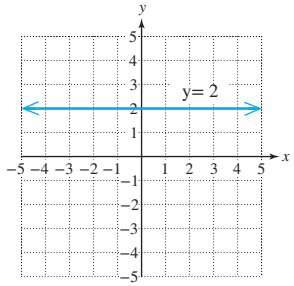
$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

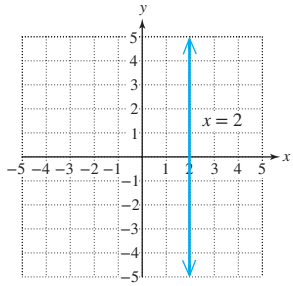
- The slope of a horizontal line is 0.
- The slope of a vertical line is undefined.

EXAMPLES

Example 1:







Example 2:
Given $(-4, 5)$ and $(6, 3)$,

(x_1, y_1) and (x_2, y_2) Label the points.

$$m = \frac{3-5}{6-(-4)} = \frac{-2}{10} = -\frac{1}{5}$$

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Slope-intercept form of a line:
Given a line with slope m and y-intercept $(0, b)$, the **slope-intercept form** of the line is given by

$$y = mx + b$$

- A function defined by $f(x) = mx + b$ $(m \neq 0)$ is a **linear function** (graph is a slanted line).
- A function defined by $f(x) = b$ is a **constant function** (graph is a horizontal line).

Average rate of change:
If f is defined on the interval $[x_1, x_2]$, then the **average rate of change** of f on the interval $[x_1, x_2]$ is the slope of the secant line containing $(x_1, f(x_1))$ and $(x_2, f(x_2))$ and is given by

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Solving equations and inequalities graphically:
The x-coordinates of the points of intersection between the graphs of $Y_1 = f(x)$ and $Y_2 = g(x)$ are the solutions to the equation $Y_1 = Y_2$.

Example 3:
Given $2x + 9y = 18$, write the equation in slope-intercept form.

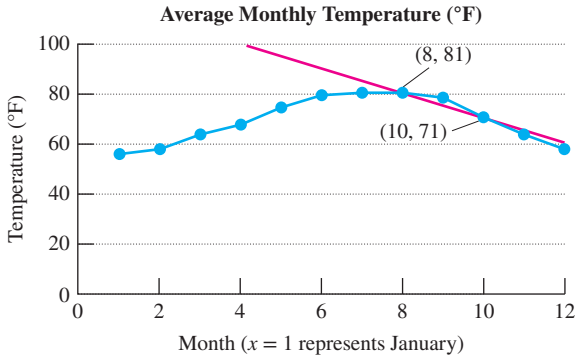
$$\begin{aligned} 2x + 9y &= 18 && \text{Solve for } y. \\ 9y &= -2x + 18 \end{aligned}$$

$$y = -\frac{2}{9}x + 2 \quad \text{or} \quad f(x) = -\frac{2}{9}x + 2$$

The slope is $-\frac{2}{9}$. The y-intercept is $(0, 2)$.

Example 4:
The average rate of change from $x_1 = 8$ to $x_2 = 10$ is

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{71 - 81}{10 - 8} = \frac{-10}{2} = -5$$



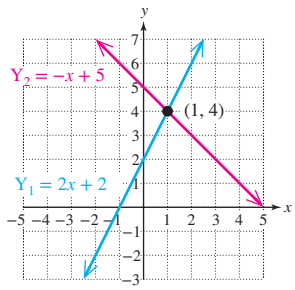
The average monthly temperature between August and October decreased by an average of 5°F per month.

Example 5:
Solve the equation and inequalities graphically.

a. $2x + 2 = -x + 5$

b. $2x + 2 < -x + 5$

c. $2x + 2 \geq -x + 5$



Solutions:

a. $\{1\}$

b. $(-\infty, 1)$

c. $[1, \infty)$

SECTION 2.5 Applications of Linear Equations and Modeling

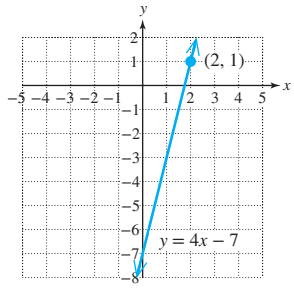
KEY CONCEPTS

The **point-slope formula** for a line is given by $y - y_1 = m(x - x_1)$, where m is the slope of the line and (x_1, y_1) is a point on the line.

EXAMPLES

Example 1:
Use the point-slope formula to find an equation of the line passing through the point $(2, 1)$ and having a slope of 4.

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 1 &= 4(x - 2) \\ y - 1 &= 4x - 8 \\ y &= 4x - 7 \end{aligned}$$



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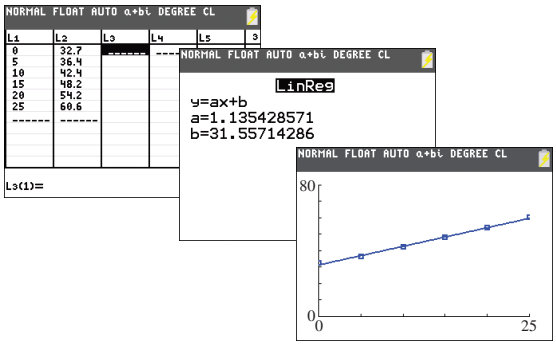
Slopes of parallel and perpendicular lines:

- If m_1 and m_2 represent the slopes of two nonvertical parallel lines, then $m_1 = m_2$.
- If m_1 and m_2 represent the slopes of two nonvertical perpendicular lines, then $m_1 = -\frac{1}{m_2}$ or equivalently $m_1 m_2 = -1$.

Applications of linear equations in two variables:

In many-day-to-day applications, two variables are related linearly.

- A linear model can be made from two data points that represent the general trend of the data.
- Alternatively, the least-squares regression line is a model that utilizes *all* observed data points. From Example 3 we have



Least-squares regression line: $y = 1.14x + 31.6$

Linear cost, revenue, and profit functions:

A **linear cost function** models the cost $C(x)$ to produce x items.

$$C(x) = mx + b$$

where m is the variable cost per item, and b is the fixed cost.

A **linear revenue function** models the revenue $R(x)$ for selling x items.

$$R(x) = px$$

where p is the price per item.

A **profit function** models the profit for producing and selling x items.

$$P(x) = R(x) - C(x)$$

Profit is revenue minus cost. The break-even point is where profit is zero (that is, where revenue equals cost).

Example 2:

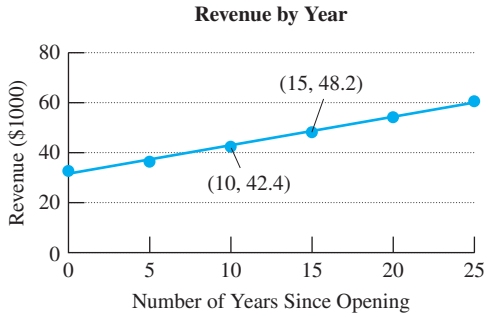
The slope of a line is $-\frac{1}{5}$.

- The slope of a line parallel to this line is $-\frac{1}{5}$.
- The slope of a line perpendicular to this line is 5.

Example 3:

The yearly revenue y (in \$1000) for a small business has increased since the year the business opened. Use the points (10, 42.4) and (15, 48.2) to write a linear model that approximates the revenue as a function of the number of years x since the business opened.

x	0	5	10	15	20	25
y	32.7	36.4	42.4	48.2	54.2	60.6



$$m = \frac{48.2 - 42.4}{15 - 10} = 1.16$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 42.4 &= 1.16(x - 10) \\ y &= 1.16x + 30.8 \end{aligned}$$

Example 4:

An herbal tea company sells tea for \$3.50 per box. The tea and packaging materials cost \$0.60 per box, and the fixed monthly costs (labor, rent, and so on) are \$3800 per month.

The cost to produce x boxes of tea during a given month is

$$C(x) = 0.60x + 3800$$

The revenue for selling x boxes of tea during a given month is

$$R(x) = 3.50x$$

The profit for producing and selling x boxes during a given month is

$$\begin{aligned} P(x) &= 3.50x - (0.60x + 3800) \\ &= 2.90x - 3800 \end{aligned}$$

To make a profit,

$$\begin{aligned} P(x) &> 0 \\ 2.90x - 3800 &> 0 \\ 2.90x &> 3800 \\ x &> 1310 \end{aligned}$$

The company will make a profit if more than 1310 boxes of tea are sold.

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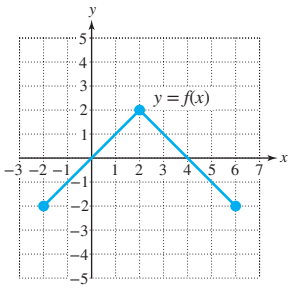
SECTION 2.6 Transformations of Graphs

KEY CONCEPTS

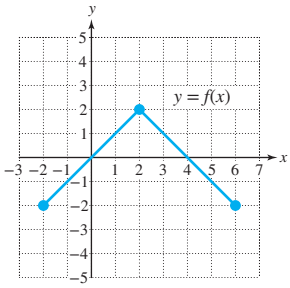
Transformations of graphs:
Consider a function defined by $y = f(x)$. Let k, h, and a represent positive real numbers. The graphs of the following functions are related to $y = f(x)$ as follows:

- Vertical translation (shift)
  - $y = f(x) + k$ Shift upward
  - $y = f(x) - k$ Shift downward
- Horizontal translation (shift)
  - $y = f(x - h)$ Shift to the right
  - $y = f(x + h)$ Shift to the left

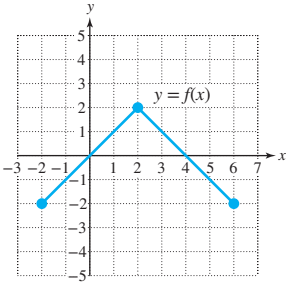
- Vertical stretch/shrink
  - $y = af(x)$ Vertical stretch (if $a > 1$)
  - Vertical shrink (if $0 < a < 1$)



- Horizontal stretch/shrink
  - $y = f(ax)$ Horizontal shrink (if $a > 1$)
  - Horizontal stretch (if $0 < a < 1$)

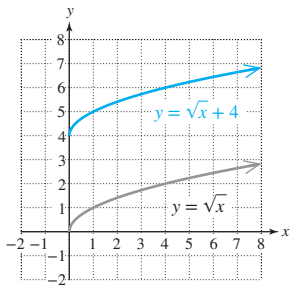


- Reflection
  - $y = -f(x)$ Reflection across the x-axis
  - $y = f(-x)$ Reflection across the y-axis

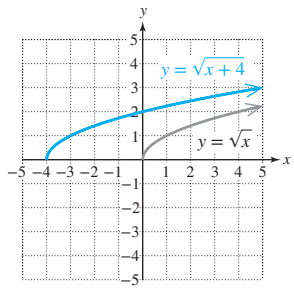


EXAMPLES

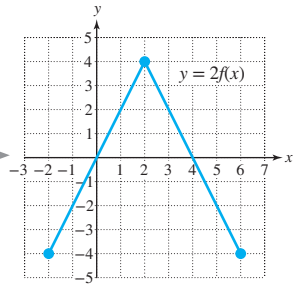
Example 1:

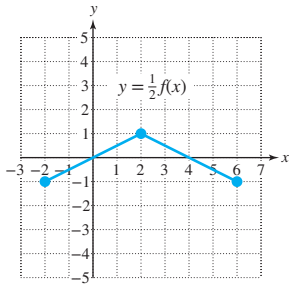


Example 2:

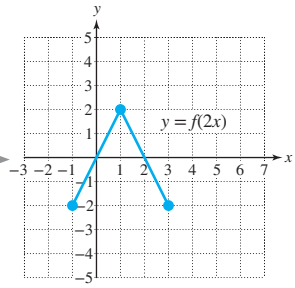


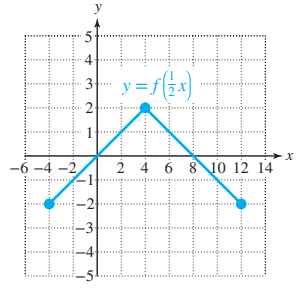
Example 3:



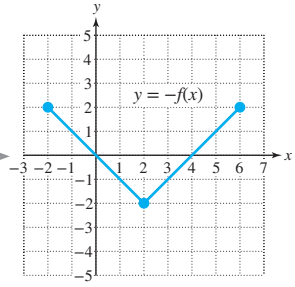


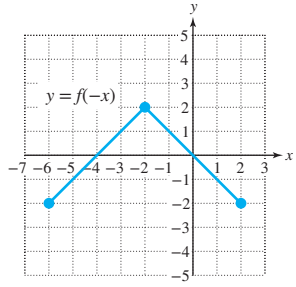
Example 4:





Example 5:





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Steps for graphing multiple transformations:

1. Horizontal translation (shift)
2. Horizontal and vertical stretch and shrink
3. Reflections across the x- and y-axes
4. Vertical translation (shift)

Example 6:

To graph $f(x) = 2\sqrt[3]{-x - 1} + 5$, perform the following transformations on the graph of $y = \sqrt[3]{x}$ in the following order.

1. Shift the graph to the right 1 unit.
2. Stretch the graph vertically by a factor of 2.
3. Reflect the graph across the y-axis.
4. Shift the graph 5 units upward.

SECTION 2.7 Analyzing Graphs of Functions and Piecewise-Defined Functions

KEY CONCEPTS

Testing for symmetry:

Consider the graph of an equation in x and y.

- The graph of the equation is symmetric to the y-axis if substituting $-x$ for x results in an equivalent equation.
- The graph of the equation is symmetric to the x-axis if substituting $-y$ for y results in an equivalent equation.
- The graph of the equation is symmetric to the origin if substituting $-x$ for x and $-y$ for y results in an equivalent equation.

Even and odd functions:

- f is an even function if $f(-x) = f(x)$ for all x in the domain of f.
- f is an odd function if $f(-x) = -f(x)$ for all x in the domain of f.

Graphing a piecewise-defined function:

To graph a piecewise-defined function, graph each individual function on its domain.

The **greatest integer function,** denoted by $f(x) = [\![x]\!]$ or $f(x) = \text{int}(x)$ or $f(x) = \text{floor}(x)$ defines $f(x)$ as the greatest integer less than or equal to x.

Increasing, decreasing, and constant behavior:

Suppose that I is an interval contained within the domain of a function f.

- f is increasing on I if $f(x_1) < f(x_2)$ for all $x_1 < x_2$ on I.
- f is decreasing on I if $f(x_1) > f(x_2)$ for all $x_1 < x_2$ on I.
- f is constant on I if $f(x_1) = f(x_2)$ for all x_1 and x_2 on I.

EXAMPLES

Example 1:

$4x^2 + 9y^2 = 1$ is symmetric to the *y-axis* because $4(-x)^2 + 9y^2 = 1$ is equivalent to $4x^2 + 9y^2 = 1$.

$4x^2 + 9y^2 = 1$ is symmetric to the *x-axis* because $4x^2 + 9(-y)^2 = 1$ is equivalent to $4x^2 + 9y^2 = 1$.

$4x^2 + 9y^2 = 1$ is symmetric to the *origin* because $4(-x)^2 + 9(-y)^2 = 1$ is equivalent to $4x^2 + 9y^2 = 1$.

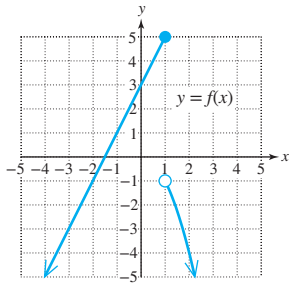
Example 2:

$f(x) = 2|x|$ ← same
$f(-x) = 2|-x| = 2|x|$ ← same — The function f is even.

$g(x) = x^3 - 5x$ — The function g is odd.
$-g(x) = -(x^3 - 5x) = -x^3 + 5x$ ← same
$g(-x) = (-x)^3 - 5(-x) = -x^3 + 5x$ ← same

Example 3:

$$f(x) = \begin{cases} 2x + 3 & \text{for } x \le 1 \\ -x^2 & \text{for } x > 1 \end{cases}$$



Example 4:

Given $f(x) = [\![x]\!]$,

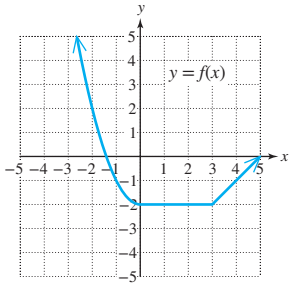
$f(2.4) = [\![2.4]\!] = 2$ and $f(-2.4) = [\![-2.4]\!] = -3$

Example 5:

f is increasing on $(3, \infty)$.

f is decreasing on $(-\infty, 0)$.

f is constant on $(0, 3)$.



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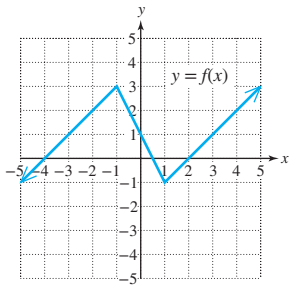
Determining relative maxima and minima:

- $f(a)$ is a **relative maximum** of f if there exists an open interval containing a such that $f(a) \geq f(x)$ for all x in the interval.
- $f(b)$ is a **relative minimum** of f if there exists an open interval containing b such that $f(b) \leq f(x)$ for all x in the interval.

Example 6:

f has a relative maximum of 3 at $x = -1$.

f has a relative minimum of -1 at $x = 1$.



SECTION 2.8 Algebra of Functions and Function Composition

KEY CONCEPTS

Operations on functions:

Given functions f and g, the functions $f + g$, $f - g$, $f \cdot g$, and $\frac{f}{g}$ are defined by

$$(f+g)(x) = f(x) + g(x)$$
$$(f-g)(x) = f(x) - g(x)$$
$$(f \cdot g)(x) = f(x) \cdot g(x)$$
$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ provided that } g(x) \neq 0$$

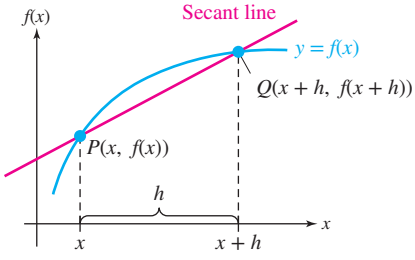
The **composition of f and g,** denoted $f \circ g$, is defined by $(f \circ g)(x) = f(g(x))$.

The domain of $f \circ g$ is the set of real numbers x in the domain of g such that $g(x)$ is in the domain of f.

Evaluate a difference quotient:

The **difference quotient** represents the average rate of change of a function f between two points $(x, f(x))$ and $(x + h, f(x + h))$.

$\frac{f(x+h) - f(x)}{h}$ Difference quotient



EXAMPLES

Example 1:

Given $f(x) = x - 3$ and $g(x) = x^2 - 7x$

$$(f+g)(x) = f(x) + g(x) = (x-3) + (x^2 - 7x)$$
$$= x^2 - 6x - 3$$

$$\left(\frac{g}{f}\right)(x) = \frac{g(x)}{f(x)} = \frac{x^2 - 7x}{x - 3} \text{ provided } x \neq 3.$$

Example 2:

Given $f(x) = \frac{1}{x + 12}$ and $g(x) = x^2 - 7x$

$$(f \circ g)(x) = f(g(x)) = \frac{1}{(g(x)) + 12}$$

$$= \frac{1}{(x^2 - 7x) + 12} = \frac{1}{(x-3)(x-4)}$$

The domain of $f \circ g$ is $\{x | x \neq 3 \text{ and } x \neq 4\}$.

In interval notation: $(-\infty, 3) \cup (3, 4) \cup (4, \infty)$.

Example 3:

Given $f(x) = 4x^2 - 5x$ evaluate the difference quotient.

$$\frac{f(x+h) - f(x)}{h}$$

$$= \frac{\overbrace{[4(x+h)^2 - 5(x+h)]}^{f(x+h)} - \overbrace{(4x^2 - 5x)}^{f(x)}}{h}$$

$$= \frac{[4(x^2 + 2xh + h^2) - 5x - 5h] - (4x^2 - 5x)}{h}$$

$$= \frac{4x^2 + 8xh + 4h^2 - 5x - 5h - 4x^2 + 5x}{h}$$

$$= \frac{8xh + 4h^2 - 5h}{h}$$

$$= 8x + 4h - 5$$

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